CHAPTER 7

REDOX REACTIONS

CHAPTER ESSENCE AT A GLANCE:
Analyzes electron-transfer reactions, oxidation numbers, types of redox processes, and systematic half-reaction equation balancing.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

- **Oxidation-Reduction Dynamics:** Oxidation involves electron loss (O.N. increases); reduction involves electron gain (O.N. decreases).
- **Oxidation State Rules:** Free elements have zero state; oxygen is -2 (except peroxides -1, OF2 +2); hydrogen is +1 (except hydrides -1).
- **Disproportionation Reactions:** A single species in an intermediate oxidation state undergoes simultaneous self-oxidation and self-reduction.
- **Ion-Electron Balancing:** Splitting into oxidation and reduction half-reactions, balancing mass via H2O and H+, and balancing charge via electrons.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\text{Oxidation: loss of } e^- \text{ (O.N. } \uparrow\text{)}, \quad \text{Reduction: gain of } e^- \text{ (O.N. } \downarrow\text{)}$$

Electronic Definition of Redox Processes

$$2H_2O_2 \rightarrow 2H_2O + O_2 \quad (\text{Oxygen: } -1 \rightarrow -2 \text{ and } 0)$$

Classic Disproportionation Reaction

$$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$$

Permanganate Half-Reaction in Acidic Medium

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$$

Dichromate Half-Reaction in Acidic Medium

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

- **Alert:** In KO2 (potassium superoxide), oxidation number of oxygen is -0.5. In OF2, oxygen is +2.
- **Alert:** In balancing ionic redox equations, both the atom counts AND the net electrical charge must be identically balanced on both sides.
- **Alert:** A species in its highest oxidation state (e.g., HNO3, KMnO4) can only act as an oxidizing agent.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define oxidation and reduction in terms of electron transfer. Identify the oxidant and reductant in: Zn + Cu2+ -> Zn2+ + Cu. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Electronic Definitions:** Oxidation is the process involving loss of electrons (increase in oxidation number). Reduction is the process involving gain of electrons (decrease in oxidation number). **[1 Mark]**

• **Identification of Reagents:** Zn loses 2 electrons to form Zn2+, so Zn undergoes oxidation and acts as the reducing agent (reductant). Cu2+ gains 2 electrons to form Cu, so Cu2+ undergoes reduction and acts as the oxidizing agent (oxidant). **[1 Mark]**

• **Key Exam Result:** Reductant = Zn; Oxidant = Cu2+.

2. What is a disproportionation reaction? Write one balanced chemical equation showing disproportionation. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** A redox reaction in which an element in one particular oxidation state undergoes simultaneous oxidation and reduction, forming two different products in higher and lower oxidation states. **[1 Mark]**

• **Balanced Chemical Equation:** 2H2O2 -> 2H2O + O2. Here, oxygen in H2O2 is in -1 oxidation state. It is reduced to -2 in H2O and oxidized to 0 in elemental O2. **[1 Mark]**

• **Key Exam Result:** Disproportionation requires the reacting element to exist in an intermediate oxidation state.

3. Assign oxidation numbers to the underlined elements in the following species: (a) Mn in KMnO4, (b) Cr in K2Cr2O7, (c) S in H2SO4, (d) N in HNO3. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **(a) Mn in KMnO4:** (+1) + x + 4(-2) = 0 => x - 7 = 0 => Oxidation number of Mn = +7. **[0.5 Mark]**

• **(b) Cr in K2Cr2O7:** 2(+1) + 2x + 7(-2) = 0 => 2x - 12 = 0 => Oxidation number of Cr = +6. **[1 Mark]**

• **(c) S in H2SO4:** 2(+1) + x + 4(-2) = 0 => x - 6 = 0 => Oxidation number of S = +6. **[0.5 Mark]**

• **(d) N in HNO3:** (+1) + x + 3(-2) = 0 => x - 5 = 0 => Oxidation number of N = +5. **[1 Mark]**

• **Key Exam Result:** Mn = +7, Cr = +6, S = +6, N = +5.

4. Balance the following redox reaction in acidic medium using the ion-electron (half-reaction) method: MnO4- + Fe2+ -> Mn2+ + Fe3+. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Half Reactions):** Oxidation half-reaction: Fe2+ -> Fe3+ + e-. Reduction half-reaction: MnO4- -> Mn2+. **[1 Mark]**

• **Step 2 (Balancing Reduction Half-Reaction):** Add 4H2O to RHS to balance oxygen: MnO4- -> Mn2+ + 4H2O. Add 8H+ to LHS to balance hydrogen: MnO4- + 8H+ -> Mn2+ + 4H2O. Add 5e- to balance charge: MnO4- + 8H+ + 5e- -> Mn2+ + 4H2O. **[1 Mark]**

• **Step 3 (Equalizing Electrons & Adding):** Multiply oxidation half by 5: 5Fe2+ -> 5Fe3+ + 5e-. Adding both half-reactions yields the balanced equation: MnO4- + 5Fe2+ + 8H+ -> Mn2+ + 5Fe3+ + 4H2O. **[1 Mark]**

• **Key Exam Result:** Balanced equation: MnO4- + 5Fe2+ + 8H+ -> Mn2+ + 5Fe3+ + 4H2O.

$$\mathrm{MnO_4^- + 5Fe^{2+} + 8H^+ \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O}$$

Balanced Permanganate-Iron Redox Reaction

5. Balance the following redox reaction in acidic aqueous medium by oxidation number method: Cr2O7(2-) + SO3(2-) -> Cr3+ + SO4(2-). **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• **Step 1 (Oxidation Number Changes):** Cr changes from +6 in Cr2O7(2-) to +3 in Cr3+ (decrease of 3 per Cr, total decrease of 6 for Cr2). S changes from +4 in SO3(2-) to +6 in SO4(2-) (increase of 2 per S). **[1 Mark]**

• **Step 2 (Equalizing Oxidation Numbers):** Multiply SO3(2-) by 3 to balance total oxidation number change: Cr2O7(2-) + 3SO3(2-) -> 2Cr3+ + 3SO4(2-). **[1 Mark]**

• **Step 3 (Balancing Charge and Mass with H+ and H2O):** Net charge on LHS = -2 + 3(-2) = -8. Net charge on RHS = 2(+3) + 3(-2) = 0. Add 8H+ to LHS to balance charge: Cr2O7(2-) + 3SO3(2-) + 8H+ -> 2Cr3+ + 3SO4(2-) + 4H2O. **[1 Mark]**

• **Key Exam Result:** Balanced equation: Cr2O7(2-) + 3SO3(2-) + 8H+ -> 2Cr3+ + 3SO4(2-) + 4H2O.

$$\mathrm{Cr_2O_7^{2-} + 3SO_3^{2-} + 8H^+ \rightarrow 2Cr^{3+} + 3SO_4^{2-} + 4H_2O}$$

Balanced Dichromate-Sulfite Redox Equation

6. Describe the construction and working of a Galvanic cell (Daniell cell). Write the anode and cathode half-reactions and the standard cell representation. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• **Construction:** Consists of zinc rod dipped in ZnSO4 solution (anode compartment) and copper rod dipped in CuSO4 solution (cathode compartment), interconnected by an external circuit and a KCl salt bridge. **[1 Mark]**

• **Electrode Half-Reactions:** At Anode (Oxidation): Zn (s) -> Zn2+ (aq) + 2e-. At Cathode (Reduction): Cu2+ (aq) + 2e- -> Cu (s). Net reaction: Zn (s) + Cu2+ (aq) -> Zn2+ (aq) + Cu (s). **[1 Mark]**

• **Standard Cell Representation:** Zn (s) | Zn2+ (aq) || Cu2+ (aq) | Cu (s). Standard EMF E°cell = E°cathode - E°anode = 0.34 V - (-0.76 V) = +1.10 V. **[1 Mark]**

• **Key Exam Result:** Daniell cell produces electrical potential of 1.10 V under standard 1 M conditions.